

Closing the Lagrange–Rvachev Loop

Exact finite synthesis of Lagrange cardinals by shifted up-functions, perfect-reconstruction projectors, and multiresolution self-expansions

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Abstract

Let $u = \text{up}$ be Rvachev’s even C^∞ probability density supported on $[-1, 1]$, with moment-generating function

$$M(z) = \int_{\mathbb{R}} e^{zx} u(x) \, dx = \prod_{r \geq 1} \frac{\sinh(z/2^r)}{z/2^r}.$$

The current ProveIt corpus already contains a sharp Strang–Fix/Appell synthesis theorem: a polynomial of degree d is reproduced by an oversampled lattice of u -atoms as soon as the radius-to-spacing ratio has 2-adic valuation at least d , and the smallest integer ratio is 2^d . It also contains a stronger one-cell compression theorem reducing the active common-scale dictionary to $d + 2$ atoms. Those results are treated here as inputs, not as new claims.

The first purpose of this report is to specialize that machinery to Lagrange cardinal polynomials on $[-1, 1]$ in a particularly transparent form. If $\tau_0, \dots, \tau_d$ are distinct nodes, ℓ_j is the corresponding Lagrange cardinal, $m = 2^d$, and $\mathcal{D}_u = M(D)^{-1}$ is the finite Appell deconvolution operator on polynomials, then

$$\ell_j(x) = \frac{1}{m} \sum_{|k| < 2m} (\mathcal{D}_u \ell_j)(k/m) u(x - k/m), \quad -1 \leq x \leq 1.$$

This is literally a finite linear combination of *shifts of the same unscaled up-function*. Its coefficients are explicit rational Bell–Bernoulli expressions whenever the nodes are rational. For symmetric nodes they obey an exact reflection law.

The second purpose is to substitute this identity into Lagrange interpolation of u and close the loop. The result is stronger and simpler than a new approximation theorem: the entire sample–analyze–up–synthesize loop is *exactly the original Lagrange interpolation projector*. In matrix form the nodal synthesis matrix S_d and the Appell analysis matrix A_d satisfy $S_d A_d = I$, hence $A_d S_d$ is an idempotent coefficient-space projector. On functions,

$$\mathcal{R}_d A_d \mathcal{E}_d = \mathcal{I}_d.$$

Thus the loop neither improves nor worsens the interpolation error: it factors it. The Runge instability already documented in the corpus for equispaced dyadic interpolation is inherited exactly, while a stable node family inherits its convergence exactly.

For nested node sets a further algebraic structure appears. The closed-loop projectors commute and satisfy $\mathcal{I}_m \mathcal{I}_n = \mathcal{I}_{\min(m,n)}$. Their differences are pairwise-annihilating idempotent detail projectors. Consequently every convergent nested Lagrange hierarchy for u becomes a genuine self-expansion of u into finite blocks of shifted copies of u . Dyadically nested Chebyshev–Lobatto nodes give a rigorous superalgebraically convergent example. The report develops this factorization, its compressed $d + 2$ -atom counterpart, its relations to the Rvachev–Appell/Kabaya–Iri polynomials and the dyadic alias nullspace, numerical checks, corpus-relative new deductions, conjectures, and a Lean formalization roadmap.

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1 Scope, corpus audit, and a terminology point

1.1 Pinned repository boundary

The repository was inspected at commit 4544e9af6b94820402c69967be1dff3f85e43c1e. The audit used the recursive Git tree, corpus-wide GitHub code search over `*.tex`, and close reading of the task-relevant current syntheses and archived antecedents. In particular, the following strands were treated as established inputs:

- the primary Fabius/Rvachev normalization, probability model, Fourier sinc product, compact support, and exact dyadic zero multiplicities;
- the inverse-and-sampling volume, including centered Rvachev–Appell polynomials, dyadic self-sampling, geometric Lagrange/ q -Richardson filters, and the Kabaya–Iri connection;
- the dyadic-comb volume, especially the documented Runge instability of global equispaced interpolation and the stable local alternatives;
- the current *Up Polynomial Synthesis* report, especially exact polynomial reproduction, the sharp order $\nu_2(m)$, the q -integer/cyclotomic alias annihilator, and the $d + 2$ -atom endpoint basis.

Historical drafts were used as provenance only when they expose an argument or normalization more explicitly than the consolidated successor. The repository is moving quickly, so the commit above, rather than a path count, is the reproducible boundary.

1.2 What is and is not meant by “a Lagrange series”

There is no single canonical infinite basis universally called *the Lagrange polynomials*: Lagrange cardinals depend on a chosen finite node set. The inspected corpus contains several different Lagrange structures: equispaced dyadic-comb interpolants, geometric-node Lagrange weights used as q -Richardson filters, and finite exact interpolation identities. It also contains the distinct Kabaya–Iri/Rvachev Appell sequence, which is a genuine polynomial sequence but is not a Lagrange cardinal sequence.

Accordingly, the loop closure below is formulated in a way that covers every legitimate interpretation. First, each finite Lagrange cardinal is synthesized exactly by finitely many shifted up-functions. Second, an arbitrary finite Lagrange interpolant is synthesized by the same finite dictionary. Third, for any *convergent hierarchy* of Lagrange interpolants, the partial sums may be substituted safely, producing a convergent self-expansion of u . This avoids an illicit interchange of an infinite series with a rapidly growing internal atom sum.

1.3 Status labels

Polynomial reproduction by translates and its Strang–Fix interpretation are classical; the particular sharp dyadic reproduction order and the explicit Appell/Bell formulas are already in the repository. In this report:

- statements explicitly called *repository input* are not claimed new;
- the pure-shift Lagrange-cardinal specialization, projector factorization, nested projector algebra, and multiresolution self-loop are deductions from those inputs which were not found in the inspected corpus by targeted searches;
- conjectures are deliberately separated from proved statements.

No claim of priority over unpublished work is intended.

2 Rvachev normalization and the synthesis input

2.1 Fourier and moment generating functions

We use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx. \quad (2.1)$$

For $u = \text{up}$,

$$\widehat{u}(\xi) = \prod_{r=0}^{\infty} \text{sinc}_{\pi}\left(\frac{\xi}{2^r}\right), \quad \text{sinc}_{\pi}(t) = \frac{\sin(\pi t)}{\pi t}, \quad (2.2)$$

and

$$\text{ord}_{\xi=n} \widehat{u} = 1 + \nu_2(n), \quad n \in \mathbb{Z} \setminus \{0\}. \quad (2.3)$$

The moment-generating function is

$$M(z) = \int_{\mathbb{R}} e^{zx} u(x) dx = \prod_{r=1}^{\infty} \frac{\sinh(z/2^r)}{z/2^r}. \quad (2.4)$$

It is even, $M(0) = 1$, and its logarithm has the cumulant expansion

$$\log M(z) = \sum_{s \geq 1} \kappa_{2s} \frac{z^{2s}}{(2s)!}, \quad \kappa_{2s} = \frac{2^{2s-1} B_{2s}}{s(2^{2s} - 1)} = \frac{B_{2s}}{2s(1 - 4^{-s})}. \quad (2.5)$$

Thus $\kappa_2 = 1/9$, $\kappa_4 = -2/225$, and so forth.

2.2 The inverse smoothing operator

On polynomials define

$$\mathcal{D}_u := M(D)^{-1}, \quad D = \frac{d}{dx}. \quad (2.6)$$

Because sufficiently high derivatives of a polynomial vanish, this is a finite operator despite its formal product or exponential notation:

$$\mathcal{D}_u = \exp\left(-\sum_{s \geq 1} \kappa_{2s} \frac{D^{2s}}{(2s)!}\right) = 1 - \frac{D^2}{18} + \frac{31D^4}{16200} - \cdots. \quad (2.7)$$

Write

$$\frac{1}{M(z)} = \sum_{n \geq 0} \beta_n \frac{z^n}{n!}. \quad (2.8)$$

Then $\beta_{2r+1} = 0$ and

$$\beta_n = Y_n(0, -\kappa_2, 0, -\kappa_4, 0, -\kappa_6, \dots), \quad (2.9)$$

where Y_n is the complete exponential Bell polynomial. Hence, for $p \in \Pi_d$,

$$(\mathcal{D}_u p)(x) = \sum_{r=0}^{\lfloor d/2 \rfloor} \frac{\beta_{2r}}{(2r)!} p^{(2r)}(x). \quad (2.10)$$

This is the exact Bell–Bernoulli coefficient engine needed below.

2.3 Repository input: affine polynomial synthesis

The current synthesis report proves the following sharp statement. Let p be a polynomial, choose a lattice step $h > 0$, an integer ratio $m > 0$, and physical atom radius $\rho = mh$. If $\deg p \leq \nu_2(m)$, then with

$$p_\rho^\sharp = M(\rho D)^{-1}p \quad (2.11)$$

one has the global locally finite identity

$$p(x) = \frac{1}{m} \sum_{k \in \mathbb{Z}} p_\rho^\sharp(x_0 + kh) u\left(\frac{x - x_0 - kh}{\rho}\right). \quad (2.12)$$

The exact stationary reproduction degree is $\nu_2(m)$; therefore the smallest positive integer ratio that reproduces every degree- d polynomial is

$$m = 2^d. \quad (2.13)$$

These facts are repository inputs. The next section makes a new specialization whose point is that ρ can be frozen at one, so the atoms become literal translates rather than dilates.

3 Every Lagrange cardinal as finitely many literal shifts of up

3.1 General nodes

Let

$$-1 \leq \tau_0 < \tau_1 < \cdots < \tau_d \leq 1 \quad (3.1)$$

be distinct interpolation nodes. Their cardinal polynomials are

$$\ell_{j,d}(x) = \prod_{\substack{0 \leq r \leq d \\ r \neq j}} \frac{x - \tau_r}{\tau_j - \tau_r}, \quad \ell_{j,d}(\tau_i) = \delta_{ij}. \quad (3.2)$$

Theorem 3.1 (Pure-shift Rvachev representation of a Lagrange cardinal). *Put $m = 2^d$. For every $j = 0, \dots, d$ and every $-1 \leq x \leq 1$,*

$$\ell_{j,d}(x) = \frac{1}{m} \sum_{|k| < 2m} (\mathcal{D}_u \ell_{j,d})(k/m) u(x - k/m). \quad (3.3)$$

Thus a degree- d cardinal is a finite linear combination of at most $4 \cdot 2^d - 1$ shifted copies of the same unscaled up-function.

Proof. Apply the repository synthesis formula (2.12) with $p = \ell_{j,d}$, $x_0 = 0$, $\rho = 1$, $m = 2^d$, and $h = 1/m$. Since $\nu_2(m) = d$, the degree hypothesis is exact, and $p_\rho^\sharp = \mathcal{D}_u p$. This gives the global locally finite identity

$$\ell_{j,d}(x) = \frac{1}{m} \sum_{k \in \mathbb{Z}} (\mathcal{D}_u \ell_{j,d})(k/m) u(x - k/m).$$

The support of the k -th atom is $[k/m - 1, k/m + 1]$. If $|k| > 2m$, this support misses $[-1, 1]$. For $|k| = 2m$ it touches the interval at exactly one endpoint, where the atom has argument ± 1 and hence value zero. Therefore only $|k| < 2m$ contribute on the closed interval, proving (3.3). $\square$

Corollary 3.2 (Closed Bell–Bernoulli coefficient formula). *The coefficient of $u(x - k/m)$ in (3.3) is*

$$c_{j,k}^{(d)} = \frac{1}{m} \sum_{r=0}^{\lfloor d/2 \rfloor} \frac{\beta_{2r}}{(2r)!} \ell_{j,d}^{(2r)}(k/m), \quad |k| < 2m, \quad (3.4)$$

where β_{2r} is given by (2.9). No collocation system involving values of u is required to compute these coefficients.

Proof. Substitute (2.10) into (3.3). □

Corollary 3.3 (Arithmetic and reflection). *If all $\tau_j \in \mathbb{Q}$, then every $c_{j,k}^{(d)} \in \mathbb{Q}$. If the node set is symmetric, $\tau_{d-j} = -\tau_j$, then*

$$c_{d-j,-k}^{(d)} = c_{j,k}^{(d)}. \quad (3.5)$$

Proof. Rationality follows because the Lagrange polynomial has rational coefficients, all cumulants and hence all β_{2r} are rational, and k/m is rational. For symmetry, $\ell_{d-j,d}(x) = \ell_{j,d}(-x)$. The operator $\mathcal{D}_u$ contains only even derivatives and commutes with reflection. Evaluation at $-k/m$ gives (3.5). □

3.2 Why the exponential atom count is not the final word

The pure-shift formula has a conceptual advantage: the atom shape and width are fixed once and for all. Its cost is the sharp dyadic oversampling ratio $m = 2^d$. The current repository contains two mechanisms that compress this: its alias-nullspace elimination leaves a $d + 2$ -atom one-cell dictionary at a common dilation, while its varying-scale antiderivative ladder can use only linearly many atoms for a fixed interval. Those are different gauges of the same polynomial-reproduction phenomenon. We return to the $d + 2$ -atom form in [section 10](#).

4 A completely explicit quadratic example

Take the three nodes $-1, 0, 1$. The cardinals are

$$\ell_{-}(x) = \frac{x(x-1)}{2}, \quad \ell_0(x) = 1 - x^2, \quad \ell_{+}(x) = \frac{x(x+1)}{2}. \quad (4.1)$$

Here $d = 2$, $m = 4$, and $\mathcal{D}_u = 1 - D^2/18$ on quadratic polynomials. Hence

$$\mathcal{D}_u \ell_{-}(x) = \frac{x^2 - x}{2} - \frac{1}{18}, \quad (4.2)$$

$$\mathcal{D}_u \ell_0(x) = \frac{10}{9} - x^2, \quad (4.3)$$

$$\mathcal{D}_u \ell_{+}(x) = \frac{x^2 + x}{2} - \frac{1}{18}. \quad (4.4)$$

Consequently, for $-1 \leq x \leq 1$,

$$\ell_0(x) = \sum_{k=-7}^7 \frac{1}{4} \left(\frac{10}{9} - \frac{k^2}{16} \right) u\left(x - \frac{k}{4}\right). \quad (4.5)$$

The two outer cardinals have the same form with the coefficient polynomial from (4.2) or (4.4). The central coefficient row is symmetric. For example,

$$c_{0,0} = \frac{5}{18}, \quad c_{0,\pm 1} = \frac{151}{576}, \quad c_{0,\pm 2} = \frac{31}{144}, \quad c_{0,\pm 7} = -\frac{281}{576}. \quad (4.6)$$

These signs are not a defect: exact polynomial reproduction requires cancellation outside the central portion of the atom train.

5 Closing the finite Lagrange loop

5.1 Interpolation first, synthesis second

For data $y = (y_0, \dots, y_d)^T$, let

$$(L_d y)(x) = \sum_{j=0}^d y_j \ell_{j,d}(x) \quad (5.1)$$

be the polynomial interpolant. For a function f , let

$$E_d f = (f(\tau_0), \dots, f(\tau_d))^T, \quad I_d f = L_d E_d f. \quad (5.2)$$

Let

$$K_d = \{k \in \mathbb{Z} : |k| < 2m\}, \quad m = 2^d. \quad (5.3)$$

Define the analysis matrix $A_d \in \mathbb{R}^{K_d \times (d+1)}$ by

$$(A_d)_{kj} = \frac{1}{m} (\mathcal{D}_u \ell_{j,d})(k/m), \quad (5.4)$$

and the reconstruction operator

$$(R_d c)(x) = \sum_{k \in K_d} c_k u(x - k/m), \quad -1 \leq x \leq 1. \quad (5.5)$$

Theorem 5.1 (Exact factorization of the Lagrange interpolant). *For every nodal vector y ,*

$$\boxed{R_d A_d y = L_d y \quad \text{on } [-1, 1].} \quad (5.6)$$

Equivalently, for every function for which the samples make sense,

$$\boxed{R_d A_d E_d = I_d.} \quad (5.7)$$

Proof. By [theorem 3.1](#), the j -th column of A_d synthesizes precisely $\ell_{j,d}$. Multiply that identity by y_j and sum over j . $\square$

This is the cleanest form of the requested loop closure. Taking $y_j = u(\tau_j)$ gives

$$I_d u(x) = \sum_{k \in K_d} C_{d,k} u(x - k/m), \quad (5.8)$$

with

$$\boxed{C_{d,k} = \frac{1}{m} [\mathcal{D}_u (I_d u)](k/m) = \sum_{j=0}^d u(\tau_j) (A_d)_{kj}.} \quad (5.9)$$

Thus the coefficients in the self-representation are obtained by: sample u at the Lagrange nodes; form the interpolation polynomial; apply the fixed inverse moment operator; sample the resulting polynomial at a dyadic grid. There is no nonlinear solve.

5.2 Perfect reconstruction at the nodes

Define the nodal synthesis matrix

$$(S_d)_{ik} = u(\tau_i - k/m), \quad i = 0, \dots, d, \quad k \in K_d. \quad (5.10)$$

It is simply $S_d = E_d R_d$.

Corollary 5.2 (Perfect reconstruction and coefficient projector). *One has*

$$\boxed{S_d A_d = I_{d+1}}. \quad (5.11)$$

Consequently

$$\Pi_d^{\text{atom}} := A_d S_d \quad (5.12)$$

is idempotent:

$$(\Pi_d^{\text{atom}})^2 = \Pi_d^{\text{atom}}, \quad \text{rank } \Pi_d^{\text{atom}} = d + 1. \quad (5.13)$$

Proof. Evaluate (5.6) at all interpolation nodes. Since $E_d L_d = I$, this gives $S_d A_d = I$. Then

$$(A_d S_d)^2 = A_d (S_d A_d) S_d = A_d S_d.$$

The rank is $d + 1$, because A_d has a left inverse S_d . $\square$

Remark 5.3 (Filter-bank interpretation). The pair (A_d, S_d) is a finite perfect-reconstruction analysis/synthesis pair, but it is not an orthogonal transform. The analysis side is a Bernoulli–Appell deconvolution, while the synthesis side consists of samples of compactly supported Rvachev atoms. The large atom space contains many directions invisible to nodal sampling; $A_d S_d$ selects the canonical Appell-generated coefficient vector for the polynomial interpolant of those samples.

6 The decisive observation: the loop is exactly interpolation

Define the complete function-space loop

$$T_d := R_d A_d E_d. \quad (6.1)$$

By (5.7),

$$\boxed{T_d = I_d}. \quad (6.2)$$

This identity has several immediate consequences that are easy to miss if one looks only at the atom coefficients.

Theorem 6.1 (No hidden approximation error). *For every $f \in C[-1, 1]$,*

$$\|f - T_d f\|_\infty = \|f - I_d f\|_\infty. \quad (6.3)$$

Moreover T_d and I_d have exactly the same operator norm and hence the same Lebesgue constant.

Proof. The operators are equal by (6.2). Everything else follows identically, not merely up to a bound. $\square$

Corollary 6.2 (Runge instability is inherited, not cured). *If a sequence of Lagrange interpolation operators I_d diverges on u or has unbounded Lebesgue constants, then the corresponding closed Rvachev loops T_d have exactly the same divergence and constants.*

The dyadic-comb report in the repository finds dramatic Runge growth for global equispaced interpolation of the Fabius/Rvachev data. The present factorization shows that rewriting those interpolants as up-atom sums cannot regularize that instability. It is a change of representation, not a change of approximation operator.

Corollary 6.3 (Stable interpolation transfers verbatim). *If $I_d u \rightarrow u$ in a norm $\|\cdot\|_X$, then the finite shifted-up sums $T_d u$ converge to u in exactly the same norm and with exactly the same error sequence.*

This result is useful conceptually: convergence analysis may be done entirely in the familiar interpolation space, while the Rvachev representation is attached afterwards without an additional approximation estimate.

7 From a Lagrange series to a genuine self-up series

7.1 Safe partial-sum substitution

Suppose a representation of u is given through polynomial partial sums

$$P_N(x) = \sum_{n=0}^N b_n L_n(x), \quad \deg P_N \leq d_N, \quad P_N \longrightarrow u \quad (7.1)$$

in some topology on $[-1, 1]$. Here the L_n may be Lagrange/Newton hierarchical polynomials, cardinal blocks, or any other polynomial terms. Put $m_N = 2^{d_N}$. Applying [theorem 3.1](#) to the *whole partial sum* gives the exact identity

$$P_N(x) = \frac{1}{m_N} \sum_{|k| < 2m_N} (\mathcal{D}_u P_N)(k/m_N) u(x - k/m_N). \quad (7.2)$$

Therefore

$$u(x) = \lim_{N \rightarrow \infty} \frac{1}{m_N} \sum_{|k| < 2m_N} (\mathcal{D}_u P_N)(k/m_N) u(x - k/m_N), \quad (7.3)$$

with convergence in exactly the topology already known for $P_N \rightarrow u$.

Remark 7.1 (Why partial sums are preferable to a formal double series). Equation (7.3) is rigorous with no absolute-convergence hypothesis on the internal atom coefficients. Expanding every polynomial term separately and then interchanging n - and k -summations may be unjustified because the minimal pure-shift lattice ratio grows exponentially with polynomial degree. The finite-partial-sum formulation keeps the exact algebra and postpones all limits to a single, already controlled interpolation limit.

7.2 Termwise block form

If the original polynomial series converges unconditionally strongly enough to justify termwise synthesis, then each term has its own finite block. For $d_n = \deg L_n$, $m_n = 2^{d_n}$,

$$L_n(x) = \frac{1}{m_n} \sum_{|k| < 2m_n} (\mathcal{D}_u L_n)(k/m_n) u(x - k/m_n), \quad (7.4)$$

and hence formally

$$u(x) = \sum_{n \geq 0} \frac{b_n}{m_n} \sum_{|k| < 2m_n} (\mathcal{D}_u L_n)(k/m_n) u(x - k/m_n). \quad (7.5)$$

The right way to prove (7.5) in a concrete setting is to bound the norms of the synthesized polynomial blocks, not the raw coefficient ℓ^1 norm unless the latter is genuinely needed.

8 Nested interpolation gives an exact multiresolution algebra

This section contains the most useful corpus-relative new structural deduction. Let $X_0 \subset X_1 \subset X_2 \subset \dots \subset [-1, 1]$ be nested finite node sets, with $N_n = |X_n| - 1$, and let I_n denote interpolation into Π_{N_n} on X_n . The associated Rvachev loops T_n are constructed as above, so $T_n = I_n$.

Theorem 8.1 (Nested projector semilattice). *For all m, n ,*

$$T_m T_n = T_n T_m = T_{\min(m, n)}. \quad (8.1)$$

Proof. Assume $m \leq n$. First, $T_m f = I_m f$ is a polynomial of degree at most $N_m \leq N_n$, so the finer interpolation operator reproduces it exactly: $T_n T_m f = T_m f$. Conversely, $T_n f = I_n f$ agrees with f at every node of X_n , hence in particular at every node of X_m . Interpolating those coarse samples gives $T_m T_n f = T_m f$. Symmetry in m, n completes the proof. $\square$

Define the detail operators

$$\Delta_0 = T_0, \quad \Delta_n = T_n - T_{n-1} \quad (n \geq 1). \quad (8.2)$$

Corollary 8.2 (Pairwise-annihilating interpolation details). *For all $m, n \geq 0$,*

$$\boxed{\Delta_n^2 = \Delta_n, \quad \Delta_n \Delta_m = 0 \quad (n \neq m).} \quad (8.3)$$

Moreover

$$T_N = \sum_{n=0}^N \Delta_n. \quad (8.4)$$

Proof. For $n \geq 1$, use $T_n^2 = T_n$, $T_n T_{n-1} = T_{n-1} T_n = T_{n-1}$:

$$(T_n - T_{n-1})^2 = T_n - T_{n-1}.$$

For distinct levels, expand the product and apply (8.1); all terms cancel. The telescoping formula is immediate. $\square$

Remark 8.3 (An algebraic multiresolution, not an orthogonal one). The Δ_n are mutually annihilating idempotents, but generally not self-adjoint in a Hilbert-space inner product. They are algebraic detail channels. Their ranges consist of the new polynomial degrees of freedom introduced by the nodes $X_n \setminus X_{n-1}$, and each range has an exact finite up-atom realization.

Theorem 8.4 (Multiresolution self-expansion of up). *Assume $T_N u \rightarrow u$ in a Banach function space X . Then*

$$\boxed{u = \sum_{n=0}^{\infty} \Delta_n u} \quad (8.5)$$

with convergence in X . For each n , let $m_n = 2^{N_n}$. Then the detail polynomial has the exact finite representation

$$\Delta_n u(x) = \frac{1}{m_n} \sum_{|k| < 2m_n} [\mathcal{D}_u(\Delta_n u)](k/m_n) u(x - k/m_n). \quad (8.6)$$

Therefore (8.5) is a genuine series of finite shifted-up blocks.

Proof. The first statement is the telescoping identity $T_N u = \sum_{n \leq N} \Delta_n u$ followed by the assumed convergence. Each $\Delta_n u$ is a polynomial of degree at most N_n , so [theorem 3.1](#) gives (8.6). $\square$

9 A rigorous convergent choice: dyadic Chebyshev–Lobatto grids

Let

$$X_n = \left\{ \cos \frac{j\pi}{2^n} : 0 \leq j \leq 2^n \right\}. \quad (9.1)$$

These sets are nested because the level- n nodes occur as the even-indexed nodes of level $n+1$. The degree is $N_n = 2^n$.

Since $u \in C^\infty[-1, 1]$, standard Chebyshev interpolation estimates imply that, for every fixed $r > 0$,

$$\|u - T_n u\|_\infty = \mathcal{O}_r(N_n^{-r}). \quad (9.2)$$

Thus [theorem 8.4](#) supplies a superalgebraically convergent self-up series. The detail blocks obey

$$\|\Delta_n u\|_\infty \leq \|u - T_n u\|_\infty + \|u - T_{n-1} u\|_\infty, \quad (9.3)$$

so they decay superalgebraically as well.

There is, however, an important representation-cost warning. The pure-shift formula uses

$$m_n = 2^{N_n} = 2^{2^n}, \quad (9.4)$$

so its raw atom count is double-exponential in the hierarchy level. This is not a convergence problem; it is an internal representation problem. The compressed $d + 2$ -atom and varying-scale constructions in the repository are therefore not cosmetic: they are essential if the self-expansion is to become computationally competitive rather than merely exact.

10 Compressed common-scale loop: only $d + 2$ atoms

The current *Up Polynomial Synthesis* report proves a stronger one-cell fact. For $M = 2^d$, put

$$\psi_k(t) = u\left(\frac{t - k}{M}\right). \quad (10.1)$$

On $[0, 1]$, the local up-spline space generated by the $2M$ active atoms has exact dimension $d + 2$. The $d + 2$ rightmost atoms

$$\psi_k|_{[0,1]}, \quad M - d - 1 \leq k \leq M, \quad (10.2)$$

form a basis, and every $P \in \Pi_d$ has a unique representation

$$P(t) = \frac{1}{M} \sum_{k=M-d-1}^M b_k(P) u\left(\frac{t - k}{M}\right), \quad 0 \leq t \leq 1. \quad (10.3)$$

The repository gives both a recurrence elimination and an $\mathcal{O}(d^2)$ output-sensitive algorithm for the b_k , using the cyclotomic annihilator and a Bernoulli differential symbol.

Apply this with

$$\tilde{\ell}_{j,d}(t) = \ell_{j,d}(2t - 1). \quad (10.4)$$

Then, on $[-1, 1]$,

$$\boxed{\ell_{j,d}(x) = \frac{1}{M} \sum_{k=M-d-1}^M b_{j,k} u\left(\frac{x + 1 - 2k}{2M}\right).} \quad (10.5)$$

This is a linear combination of only $d + 2$ common-scale shifted atoms. It is not the same constraint as [theorem 3.1](#): here the atom is strongly dilated, whereas (3.3) insists on literal unscaled copies of u .

Corollary 10.1 (Compressed interpolation loop). *For any nodal data y , the degree- d Lagrange interpolant can be routed through the fixed $d + 2$ -atom endpoint basis. Thus the same interpolation projector I_d has both:*

1. a pure-shift factorization using at most $4 \cdot 2^d - 1$ unscaled atoms;
2. a compressed common-scale factorization using exactly $d + 2$ basis atoms.

The second factorization has one more atom than $\dim \Pi_d = d + 1$. This is not an accident: the repository proves that the complete one-cell up-spline space has dimension $d + 2$, while its polynomial subspace has codimension one. An arbitrary local coefficient vector therefore satisfies exactly one scalar polynomiality constraint. Closing the Lagrange loop selects that polynomial hyperplane.

11 Connections to Appell, Kabaya–Iri, and q -binomial structures

11.1 Lagrange versus Rvachev–Appell coordinates

The centered Rvachev–Appell polynomials in the repository are generated by

$$\frac{e^{xz}}{M(z)} = \sum_{n \geq 0} \mathcal{A}_n(x) \frac{z^n}{n!}. \quad (11.1)$$

They are the centered form of the Kabaya–Iri scaling polynomials studied in the external refinement literature. The same reciprocal $1/M$ appears in our Lagrange coefficient operator $\mathcal{D}_u$. Indeed, if $p(x) = \sum_n a_n x^n$, then

$$\mathcal{D}_u p(x) = \sum_n a_n \mathcal{A}_n(x). \quad (11.2)$$

Thus the coefficient row in (3.3) is obtained by a two-stage finite change of coordinates:

$$\text{Lagrange cardinal} \longrightarrow \text{monomials} \longrightarrow \text{Rvachev–Appell polynomials} \longrightarrow \text{dyadic samples}. \quad (11.3)$$

This makes the Bell–Bernoulli coefficients inevitable rather than ornamental.

11.2 Where the q -binomial algebra enters

The pure-shift formula uses additive dyadic centers $k/2^d$, whereas much of the repository’s q -binomial Lagrange calculus uses multiplicative geometric nodes. The bridge is not that the two node sets are identical. It is that both are controlled by binary dilation:

- geometric Lagrange denominators factor into q -Pochhammer symbols at $q = 1/2$ or $1/4$;
- the up Fourier zeros have multiplicity $1 + \nu_2(n)$, so the alias annihilator factors into dyadic q -integers;
- eliminating the exponentially many pure-shift coefficients uses precisely that cyclotomic/ q -integer annihilator.

The two appearances of Gaussian binomial algebra are therefore different faces of one binary scaling geometry.

11.3 Thue–Morse as the complementary detail character

Repeated differentiation of a coarse up-atom produces a finite signed block of finer atoms with Thue–Morse signs in the repository. The alias annihilator used to compress the pure-shift lattice is built from the same dyadic factors. The new nested projector details Δ_n suggest a third role: Thue–Morse blocks may provide natural coefficient-space coordinates for the kernels separating adjacent closed-loop levels. This is developed as a conjectural program in [section 15](#).

12 Stability has two different meanings

12.1 External stability: exactly the Lebesgue constant

As a map from sampled function values to the reconstructed function on $[-1, 1]$, the loop is simply I_d . Hence its external condition number in L^∞ is governed by the ordinary Lebesgue function

$$\lambda_d(x) = \sum_{j=0}^d |\ell_{j,d}(x)|, \quad \Lambda_d = \max_{x \in [-1, 1]} \lambda_d(x). \quad (12.1)$$

There is no additional “Rvachev penalty” in function space.

12.2 Internal stability: atom coefficients can still be large

The factorization $I_d = R_d A_d E_d$ can have large intermediate norms even if the product is well conditioned. This is familiar from nonorthogonal factorizations: cancellation may be essential. The inverse moment operator $\mathcal{D}_u$ amplifies high derivatives of high-degree cardinals, and the pure-shift channel is massively redundant. Therefore

$$\|A_d\| \|R_d\| \quad (12.2)$$

can be far larger than $\|I_d\|$. The compressed $d + 2$ -atom basis reduces dimension but may itself have large signed coefficients; the repository explicitly flags scale optimization and conditioning as open problems.

This distinction is central for implementation. If only the reconstructed function matters and coefficients are carried exactly, the factorization is benign. If coefficients are quantized, truncated, independently perturbed, or used as trainable parameters, internal conditioning becomes decisive.

13 Numerical verification

The accompanying script `rvachev_lagrange_loop_experiments.py` performs an independent numerical check of the degree-two formulas. It computes all polynomial/Appell coefficients with exact rational arithmetic. It then evaluates u from the Fourier inversion

$$u(x) = 2 \int_0^\infty \hat{u}(\xi) \cos(2\pi\xi x) \, d\xi \quad (13.1)$$

using a 50-factor sinc product and intervalwise quadrature through frequency 40. The numerical approximation is used only to test the exact theorem.

For nodes $-1, 0, 1$, $m = 4$, and the 15 nontrivial shifts $k/4$, $-7 \leq k \leq 7$, the experiment gives

Quantity	observed value
$\ S_2 A_2 - I_3\ _\infty$	6.25×10^{-10}
$\ (A_2 S_2)^2 - A_2 S_2\ _\infty$	1.21×10^{-9}
max error for ℓ_- on 65 grid points	3.24×10^{-10}
max error for ℓ_0 on 65 grid points	2.35×10^{-10}
max error for ℓ_+ on 65 grid points	3.24×10^{-10}

The eigenvalues of the numerically formed 15-by-15 coefficient projector are three values equal to one to numerical precision and twelve values equal to zero, exactly as [theorem 5.2](#) predicts. The residual scale is set by the independent Fourier quadrature, not by the rational coefficient algebra.

14 Further deductions

14.1 A universal factorization theorem for any polynomial projector

The Lagrange nature of I_d is only needed to identify a convenient evaluation map. More generally, let $P_d : C[-1, 1] \rightarrow \Pi_d$ be any linear polynomial projector. Applying the pure-shift synthesis map to $P_d f$ yields

$$P_d f(x) = \frac{1}{2^d} \sum_{|k| < 2^{d+1}} [\mathcal{D}_u(P_d f)](k/2^d) u(x - k/2^d). \quad (14.1)$$

Thus every polynomial approximation scheme—least squares, orthogonal projection, Hermite interpolation, constrained interpolation—admits an exact Rvachev-atom factorization. What is special about Lagrange interpolation is the particularly transparent perfect-reconstruction matrix $S_d A_d = I$.

14.2 Cardinal columns are exact dual certificates

Because $S_d A_d = I$, the j -th coefficient column $a^{(j)}$ satisfies

$$\sum_{k \in K_d} a_k^{(j)} u(\tau_i - k/m) = \delta_{ij}. \quad (14.2)$$

Thus each Lagrange cardinal produces an explicit finite dual certificate for the sampling matrix of shifted up-atoms. For rational nodes the certificate coefficients are rational even though the matrix entries $u(\tau_i - k/m)$ are usually transcendental or arithmetically mysterious. This separation of exact rational analysis coefficients from special-function synthesis values is striking.

14.3 Nested details provide canonical cancellation blocks

For nested nodes, $\Delta_n u$ vanishes at every old node in X_{n-1} . Its shifted-up coefficient vector therefore lies in the nullspace of the old nodal synthesis matrix after interscale transfer. The identities $\Delta_n \Delta_m = 0$ show that these nullspaces are not arbitrary: they come with a canonical hierarchy induced by polynomial degree. This is the natural place to look for a Thue–Morse/cyclotomic basis of details rather than merely a basis of the full alias kernel.

15 Conjectures and open problems

Conjecture 15.1 (Thue–Morse diagonalization of nested detail channels). *For dyadically nested node sets, there exists a choice of interlevel coefficient transfer maps for which the atom-space realization of each detail projector Δ_n decomposes into finitely many blocks generated by the repository’s q -integer annihilator $\prod_r [2^r]_E$, and whose highest-frequency signs are the corresponding Thue–Morse characters. After a natural dyadic weighting, these blocks should be uniformly better conditioned than an arbitrary basis of the same kernel.*

The exact annihilator and the exact detail algebra are both known; what remains is to identify the best common coordinates and prove a quantitative condition bound.

Conjecture 15.2 (Balanced gauge for pure-shift Chebyshev cardinals). *For Chebyshev–Lobatto cardinals, the canonical Appell coefficient vector in the large pure-shift dictionary can be altered by exact alias-nullspace vectors so that its weighted ℓ^1 norm grows subexponentially in the polynomial degree, while preserving the represented cardinal identically on $[-1, 1]$.*

The unweighted raw coefficient count is exponential and no such statement follows from function-space stability alone. A negative result would also be valuable: it would prove an intrinsic gap between the excellent external conditioning of Chebyshev interpolation and any fixed-width pure-shift realization.

Conjecture 15.3 (Condition-optimal blend of left and right compressed bases). *The left and right $d + 2$ -atom endpoint bases from the repository can be blended, using their common polynomial subspace, to produce a symmetric compressed factorization whose worst cardinal coefficient norm is asymptotically smaller than that of either one-sided endpoint basis. The optimal blend should be characterized by a one-dimensional balancing condition associated with the codimension-one nonpolynomial direction of the local up-spline space.*

Problem 15.4 (Exact change of basis to Kabaya–Iri polynomials). *For a structured node family, derive a closed determinant-free formula for the matrix changing the Lagrange cardinals into the centered Rvachev–Appell basis, and then combine Lee’s derivative biorthogonality with (14.2). The goal is a single formula relating three dual systems: point evaluations, derivatives of u , and shifted-up coefficient certificates. For rational or geometric nodes one should expect Bernoulli and Gaussian q -binomial factors to appear simultaneously.*

Problem 15.5 (Coefficient-space renormalization group). *For a nested hierarchy, construct explicit interscale maps $J_{n \rightarrow n+1}$ between the finite atom coefficient spaces so that*

$$R_{n+1}J_{n \rightarrow n+1} = R_n$$

on Π_{N_n} . Determine the spectrum and Jordan structure of the induced maps on the alias kernels. The dyadic zero multiplicities strongly suggest that these maps should admit a triangular decomposition indexed by 2-adic valuation.

Problem 15.6 (Optimal true-shift complexity). *The pure-shift construction uses $4 \cdot 2^d - 1$ shifts of a fixed-width atom on $[-1, 1]$. The repository proves $d + 2$ atoms are enough when a common dilation is allowed. Determine the smallest number N_d^{shift} of unscaled translates of u , with arbitrary centers and coefficients, whose span contains all of Π_d on $[-1, 1]$. The sharp growth rate of N_d^{shift} is presently unclear and is a cleaner invariant than the lattice-specific $4 \cdot 2^d - 1$ count.*

16 Promising computational directions

Several computations can push the theory without requiring heroic symbolic manipulation.

1. **Exact rational cardinal tables.** For rational symmetric nodes, generate A_d exactly from (3.4); quotient its columns by the known cyclotomic alias nullspace; compare ℓ^1 , ℓ^2 , and max norms before and after optimization.
2. **Chebyshev algebraic tables.** Work over the real cyclotomic field containing $\cos(j\pi/N)$ or use high-precision ball arithmetic; compare external Lebesgue growth with internal coefficient growth.
3. **Nested detail singular vectors.** Numerically compute the interlevel coefficient maps, then correlate their singular vectors with Thue–Morse and 2-adic valuation patterns.
4. **Compressed-basis balancing.** Construct left/right endpoint bases and solve the one-parameter minimax blend suggested by theorem 15.3.
5. **Certified Fourier validation.** Replace floating quadrature in the supplied script by interval arithmetic using the corpus’s sharp Fourier envelope, producing rigorous enclosures for $S_d A_d - I$ at moderate degrees.

17 Lean formalization roadmap

Most of the new algebra should be substantially easier to formalize than the analytic synthesis theorem on which it rests.

17.1 Finite Lagrange layer

Define a finite node vector, the cardinal polynomials, and an evaluation linear map. Mathlib already has polynomial interpolation infrastructure, but the concrete cardinal product (3.2) is useful because the target theorem needs explicit evaluation and reflection properties.

17.2 Bridge to the existing Appell synthesis API

A theorem corresponding to theorem 3.1 should be a short specialization of the existing polynomial-synthesis declaration with $\rho = 1$, $h = 2^{-d}$, and interval $[-1, 1]$, followed by support pruning. The Bell coefficient corollary is finite polynomial algebra.

17.3 Matrix perfect reconstruction

After defining the finite index set K_d , package A_d and S_d as matrices over $\mathbb{R}$. The proof of $S_d A_d = 1$ is pointwise cardinal evaluation. Idempotence of $A_d S_d$ is then one line of matrix algebra.

17.4 Nested projector algebra

Define interpolation projectors T_n abstractly by their reproduction and nodal properties. For nested node sets, prove (8.1) without mentioning up-functions. The detail identities follow by ring normalization. Finally rewrite T_n through the Rvachev factorization. This separation should make the Lean proof robust to future changes in the analytic implementation of u .

18 Conclusions

The requested loop does close, but its most informative form is not a mysterious functional fixed-point equation. It is an exact factorization theorem.

For degree d , every Lagrange cardinal on $[-1, 1]$ has the explicit pure-shift representation

$$\ell_{j,d}(x) = 2^{-d} \sum_{|k| < 2^{d+1}} (M(D)^{-1} \ell_{j,d})(k/2^d) u(x - k/2^d),$$

with finite Bell–Bernoulli coefficients. Substituting these cardinals into the Lagrange interpolant of u yields a finite self-representation of the interpolant. The sample–analysis–synthesis matrices satisfy $S_d A_d = I$, and the complete function loop is exactly I_d . This identity settles the first-order analysis: all approximation error, convergence, Lebesgue growth, and Runge instability are those of the chosen interpolation scheme and no others.

The genuinely new frontier appears when levels are composed. On nested grids the closed loops form commuting projectors with $T_m T_n = T_{\min(m,n)}$, and their differences are pairwise-annihilating detail projectors. Any convergent nested interpolation scheme therefore produces a convergent self-expansion of u into finite Rvachev-atom blocks. Dyadic Chebyshev–Lobatto grids give a rigorous superalgebraically convergent example; the repository’s $d + 2$ -atom endpoint basis and varying-scale ladders offer routes to compress its otherwise enormous pure-shift realization.

The next technical problem is no longer existence. It is to find the best coefficient-space gauge: one that combines the exact cyclotomic/Thue–Morse nullspace, stable interpolation nodes, and the Rvachev–Appell duality into a well-conditioned multiresolution transform. That problem sits exactly at the intersection of the corpus’s sinc-zero arithmetic, q -binomial interpolation, Bell–Bernoulli deconvolution, and atomic-function synthesis.

A Exact coefficient recipe

For reproducibility, here is the direct symbolic algorithm for one Lagrange cardinal.

Algorithm A.1 (Pure-shift Lagrange cardinal). Input: distinct nodes $\tau_0, \dots, \tau_d \subset [-1, 1]$ and index j .

1. Form

$$\ell_{j,d}(x) = \prod_{r \neq j} \frac{x - \tau_r}{\tau_j - \tau_r}.$$

2. Set $m = 2^d$.

3. Compute cumulants κ_{2s} for $1 \leq s \leq \lfloor d/2 \rfloor$ from (2.5).

4. Compute $\beta_0, \beta_2, \dots$ from $1/M(z)$ or the complete Bell formula (2.9).

5. Form the finite polynomial

$$q_j(x) = \sum_{s=0}^{\lfloor d/2 \rfloor} \frac{\beta_{2s}}{(2s)!} \ell_{j,d}^{(2s)}(x).$$

6. For each integer $-2m < k < 2m$, emit the atom

$$\frac{q_j(k/m)}{m} u(x - k/m).$$

The sum is exactly $\ell_{j,d}$ throughout $[-1, 1]$.

B Reproduction information from the repository

The central source for the synthesis input is the current file

Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/representations/Up_Polynomial_Synthesis/Up_Polynomial_Synthesis.tex.

At the pinned commit it contains, among other results: the exact stationary reproduction order $\nu_2(m)$; the canonical minimum $m = 2^d$; Appell moment deconvolution; the complete dyadic alias nullspace generated by a product of dyadic q -integers; the exact one-cell dimension $d + 2$; the unique right endpoint basis; an $\mathcal{O}(d^2)$ output-sensitive coefficient algorithm; and a Thue–Morse derivative stencil. The present report uses these as inputs and adds the Lagrange-cardinal and projector structures.

The main interpolation source is the current dyadic-comb frontier volume, which contains global Lagrange interpolation on additive dyadic nodes and documents the Runge transition. The inverse-and-sampling volume contains the centered Rvachev–Appell/Kabaya–Iri theory and geometric Lagrange q -Richardson filters.

C Literature context

Polynomial reproduction from shifts of a generator belongs to the classical Strang–Fix framework. Continuous refinement operators and the up-function were studied by Rvachev and, independently, by Kabaya and Iri. Lee’s scaling-polynomial work places the Kabaya–Iri polynomials in a general Appell/biorthogonal framework for refinable functions. The present factorization is compatible with that literature but asks a different question: not which polynomial sequence is biorthogonal to derivatives of u , but how a prescribed finite interpolation projector can be routed exactly through finite up-atom synthesis.

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